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Using Large Language Models to Assist Antimicrobial Resistance Policy Development: Integrating the Environment into Health Protection Planning

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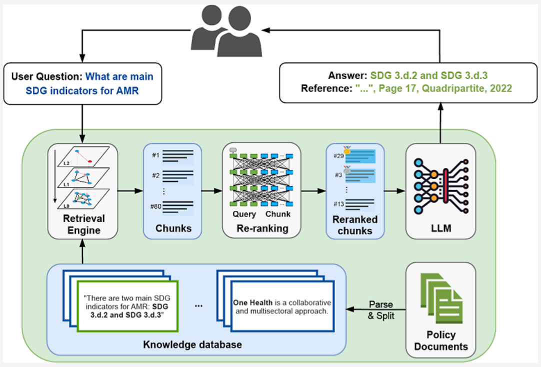
ABSTRACT: Increasing antimicrobial resistance (AMR) poses a substantial threat to global health and economies, which has led many countries and regions to develop AMR National Action Plans (NAPs). However, inadequate logistical capacity, funding, and essential information can hinder NAP policymaking, especially in low-to-middle-income countries (LMICs). Therefore, major gaps exist between aspirations and actions, such as fully operationalized environmental AMR surveillance programs in NAPs. To help bridge knowledge gaps, we compiled a multilingual database that contains policy guidance from 146 countries composed of NAPs, internal reports, and other guidance documents on AMR mitigations, including environmental considerations. Leveraging this database, we developed an AMR-Policy GPT, a large language model with advanced retrieval-augmented generation capabilities. This prototype model can search and summarize evidence from plans, metadata, and technical knowledge and provide traceable references from global document databases. It was then manually validated to show its proficiency in accurately managing diverse inquiries while minimizing misinformation. Overall, the AMR-Policy GPT offers a prototype that, with the deepening of its database and further road testing, has the potential to support inclusive, evidence-informed AMR policy guidance to support governments, research, and public agencies. A conversational version of our prototype is available at www.liuhuibot.com/amrpolicy.



KEYWORDS: *antimicrobial resistance, policymaking, one health, large language model, artificial intelligence, retrieval-augmented generation, LMICs*

1. INTRODUCTION

Increasing levels of antimicrobial resistance (AMR) is a global health crisis that impacts humans, animals, plants, and the wider environment.[1] In 2019, an estimated 4.95 million deaths were linked to bacterial antibiotic resistance, including 1.27 million deaths directly attributed to AMR infections.[2] AMR has led to increased healthcare costs, less effective treatments, extended hospital stays, and higher mortality rates, particularly in low-to-middle-income countries (LMICs) where environmental quality is often poor and the burden of AMR is most severe.[2] It is estimated that over 50% of resistant human infections in LMICs result from contaminated food or water,[3] although surveillance of possible environmental AMR exposures is rarely performed.

With the need for global consensus and coordinated efforts to tackle AMR, the World Health Organization (WHO) formulated a Global Action Plan (GAP) in 2015,[4] with 194 WHO member states committing to developing country-specific AMR National Action Plans (NAPs).[5] In November 2023, 178 countries formulated multisectoral NAPs, with some countries such as Australia, China, and the UK progressing to implement second iterations.[6,7] Associated with NAPs, the Global Antimicrobial Resistance and Use Surveillance System (GLASS) was launched in 2015 to standardize AMR surveillance,[8] which now includes 127 countries.[9]

Although NAPs and data from surveillance systems are key to addressing global AMR, problems exist. For example, Patel et al.[7] reviewed NAPs from around the world and found that plan implementation was limited due to inadequate financing and incomplete surveillance plans. Another assessment of AMR NAPs for human health emphasized the need for adaptability to varying resource constraints in optimal medication use.[10] In general, NAPs often lack specificity to local contexts and do not truly follow One Health principles, such as not considering

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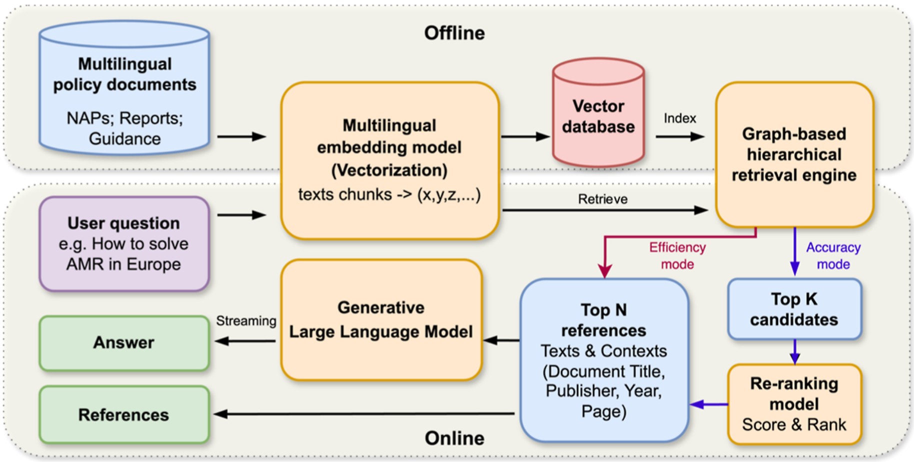

Figure 1. Schematics of AMR-Policy GPT workflow.

different data needs among sectors in surveillance or drawing lessons from similar countries. These problems are primarily attributed to information, communication, and education gaps, which are most acute in the LMICs. LMICs encounter greater challenges because they often lack environmental protection and/or surveillance networks that cross-sectors, even though they are at greater risk of AMR transmission and spread due to poor civil infrastructure.[11] Therefore, LMICs require locally relevant evidence and information to establish clearer objectives in their own context for both evaluating policy effectiveness and cost-effective implementation.

As such, comprehensive summaries of AMR actions and consequences across different sectors—including historical and existing plans, as well as surveillance data and reports from environmental sectors, and even lessons learned during the COVID-19 pandemic[9,12]—would help guide policy decisions by drawing insights from various countries and organizations (e.g., GLASS). However, our rapidly expanding knowledge of AMR under different One Health sectors and the frequent updates of related documents, reports, and NAPs require significant resources to stay up-to-date on relevant information, suggesting the need for a central database and an effective tool to support policy decisions.

A promising option for information extraction and evidence integration is using artificial intelligence-based large language models (LLMs), which have shown remarkable proficiency in natural language understanding and generation tasks.[13−15] Although LLMs have great value, they encounter challenges in answering factual questions due to limitations in the root information, such as hallucinations or confabulations, as well as reliance on outdated and static knowledge or missing information.[16−18] To overcome such limitations, retrieval-augmented generation (RAG) methods have been introduced to extend LLMs' capabilities by incorporating external knowledge sources.[19] By using information retrieval strategies, RAG empowers LLMs to procure and assimilate relevant data from designated knowledge bases, thereby improving more factual and accurate outputs of LLMs.[19,20] The utilization of RAG broadens the utility of LLMs across a spectrum of applications, ranging from document searching and analysis to integration and summarization.[21,22]

Here, we collected AMR-related policy documents from 146 countries encompassing NAPs, reports, and submissions to international organizations. These documents acted as an "expert" knowledge database on which we developed a prototype RAG-based conversational LLM called the AMR-Policy GPT. The relevance of such data for local, evidence-based policymaking, of course, depends on whether the recruited evidence is being applied to a similar context. Given that evidence may be limited in the resource-constrained contexts where this is most needed, sourcing such data by clear citation allows evidence synthesized by the AMR-Policy GPT to be used more accurately and effectively by local policymakers. Therefore, our objective here is to provide a user-friendly prototype tool that could recruit current and contextually relevant information on AMR. Although a prototype is presented here, we hope the AMR-Policy GPT might evolve into a tool that can support evidence-based AMR policymaking by increasing knowledge sharing across countries and sectors worldwide.

2. METHODS

2.1. Model Development—Definition of Retrieval-Augmented Generation. LLMs, like "GPT-4" (i.e., ChatGPT), are trained on extensive training data and use billions of parameters to perform tasks, such as question answering.[17] However, LLMs require continual training; therefore, we introduce a RAG that allows our LLM to acquire domain-specific capabilities without retraining, a more cost-effective approach to optimizing LLM outputs. Specifically, the term RAG refers to a technology that expands the capabilities of LLMs. RAG is used to redirect the LLM to retrieve relevant information from predetermined knowledge sources in response to queries, thus enabling LLMs to generate more accurate and credible queries. This empowers the LLMs to exercise control over the generated text output while providing users with traceable and trustful references.[22,23] In parallel, the external databases can be updated frequently, which provides high scalability and flexibility under the increasing documents. As an example, Vaghefi et al.[22] developed ChatClimate, a RAG-based LLM that integrates the external knowledge resource of the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC AR6) into GPT-4, creating a more up-to-date and reliable question-answering system for climate change.

To distill down the growing volume and variable quality of AMR guidance and planning documents, a RAG-based LLM was developed here, called the AMR-Policy GPT, which has an interactive and user-friendly chatbot interface. This chatbot could transform information from the dataset using a dialogue

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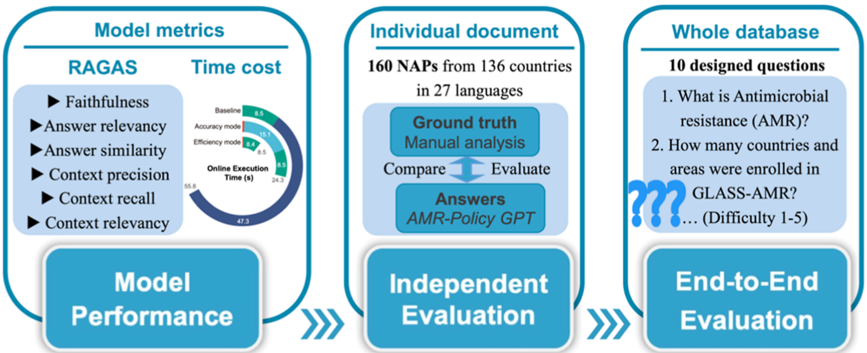

Figure 2. Evaluation framework. After automatically evaluating model performance by RAGAS framework, independent evaluation and end-to-end evaluation were conducted. Independent evaluation focused on assessing the model's accuracy in conducting searches and interpreting information from specific documents. End-to-end evaluation evaluated the capacity of AMR-Policy GPT to search, gather, and integrate information across the entire database, enabling it to answer broader questions.

format, creating a tool for recruiting evidence for AMR policymaking that can translate and share global knowledge to answer user's questions. It can address a wide range of queries about the status of AMR, NAPs, and policymaking across countries. As an example, users can query specific definitions of AMR-related knowledge, obtain the latest AMR evidence based on available data, access statistical data on topics like the proportion of drug resistance in pathogens, and even make between-country comparisons on governmental and other responses to AMR. With the capability of the RAG framework to integrate updated external databases into the LLM, AMR-Policy GPT can remain up-to-date, catering to users' requirements for accessing and querying the most recent information on AMR policy.

Overall, the mechanism of the RAG framework can be summarized as follows

$$P(A|Q) = \sum_{i=1}^{k} P(A|D_i, Q) \times P(D_i|Q)$$

Here, Q represents the query posed by the user and D represents the most pertinent set of text chunks $D_1,\cdots, D_k$ to Q from the collection of AMR policy documents (E). During information retrieval, the most relevant text chunks $D_1,\cdots, D_k$ are collected from (E) to the query (Q) posed by the user. This process is summarized as $P(D|Q)$, which represents the probability of selecting documents D from E, contingent upon query Q, which is then calculated by the retrieval engine. After the retrieval phase, a transformer-based text generator produces an appropriate response (A) based on pertinent retrieved text chunks (D). The generation phase is illustrated as $P(A|D,Q)$, representing the probability distribution for generating answer A, given the set of retrieved text chunks D and the user's query Q. $P(A|D)$ represents the possibilities.

2.2. Pipeline for the AMR-Policy GPT Tool. The pipeline of AMR-Policy GPT is shown in Figure 1, which contains two sections: an offline index and an online service. The offline section includes dataset preparation and construction of the vector database. The online section is initiated with the question of the user and proceeds to a multilingual embedding model, with the retrieval engine retrieving the most related content (e.g., top K candidates). Two search modes exist: the quicker, less exact "efficiency" approach and the slower but more precise "accuracy" approach. Both approaches generate the top N references to the generative LLM. However, the accuracy mode goes further, reranking references based on a score calculated by a reranking model. The greater the relevance between the query string and the reference sentences, the higher the score. The accuracy mode, however, can cost more runtime.

After retrieval, final references are embedded into a predesigned, engineered prompt platform. Queries are generated by the generative LLM GPT-4 by OpenAI API and shown on our interactive webpage (accessible at www.liuhuibot.com/amrpolicy). Simply, the user's queries are entered on the web page, and answers and related references are returned to the user. Users also have two choices relative to the source of the generated results: the standalone mode (which provides answers based solely on our database) and the hybrid mode (answers from our database plus in-house knowledge of GPT-4).

2.3. Dataset Preparation. To create an external knowledge base for the RAG framework, 383 publicly available government policy documents related to combating AMR were obtained from 146 different countries (detailed in Supporting Information) using PDF formats. This database encompasses most countries that have declared their participation in the GAP, providing the most up-to-date and comprehensive range of documents available. The database contains three types of documents: NAPs, reports (e.g., midterm reports and situation analyses), and guidelines. At the time of the report gathering, there were 160 NAPs across countries, four action plans from intergovernmental organizations (e.g., the GAP on AMR, the FAO Action Plan on AMR 2016−2020, and the European One Health Action Plan against AMR), 58 guidance documents, and 161 reports.

We used pdfplumber,[24] a PDF parsing tool, to extract and transform data and information from PDFs. The PDF documents were parsed, generating parsing files that contained information from the documents for processing and analysis. To ensure semantic integrity and readability of textual data while retaining maximum information, the AMR documents were segmented into smaller chunks. Detailed information, such as textual content, document title and type, publication year, page number, pertinent countries, and publisher, was stored for each chunk.

2.4. Construction of Vector Database, Retrieval Engine, and Reranking Model. For the vectorization component, a multilingual embedding model was used, i.e., BAAI/BGE-M3,[25] to allow both the offline vectorization of chunks and the online vectorization of user queries. This model is proficient in supporting over 100 languages and performs

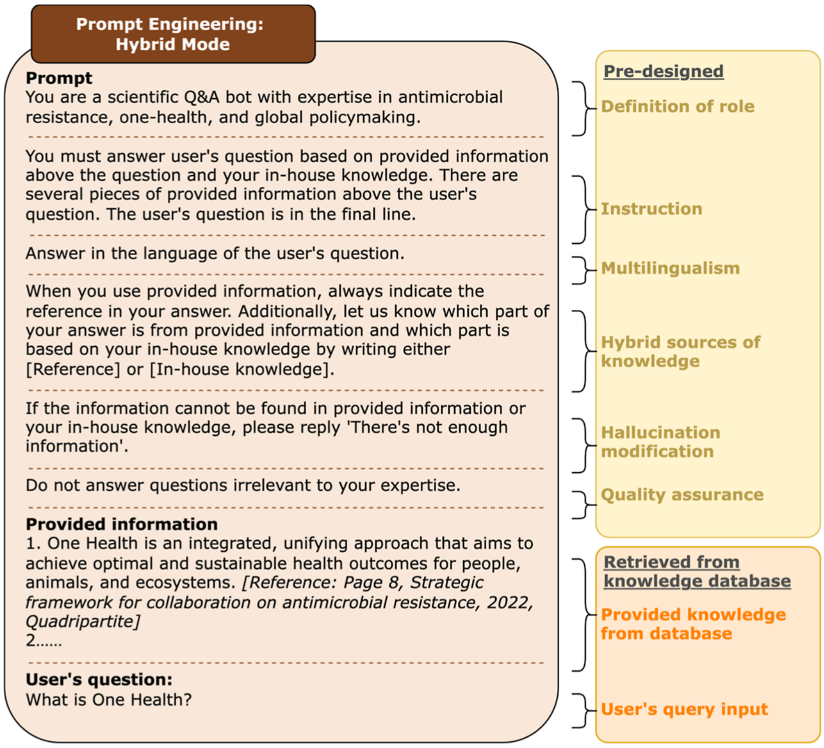

Figure 3. Illustration of prompt engineering in the hybrid mode of AMR-Policy GPT and the corresponding design principles. Left side depicts the detailed prompt measures implemented in the hybrid mode, while the right side presents the designed principles. Prompt serves as a preguide for AMR-Policy GPT, allowing it to focus on answering AMR-relevant questions and providing references through the RAG system.

effectively in embedding text segments into 1024-dimensional vectors. This capability enables the model to handle multilingual inquiries and search for results in documents written in multiple languages.

A vector database was then devised for the retrieval and storage of embeddings, along with associated contextual information. The database was loaded onto servers and deployed for computation and offline serialization. In parallel, a graph-based hierarchical reference retrieval engine, based on the Hierarchical Navigable Small World (HNSW) algorithm, was integrated into the vector database,[26] enhancing the efficiency of high-dimensional reference data retrieval. The engine was developed to retrieve the top-K nearest embeddings in a vector database compared with the query embedding using HNSW. Additionally, for the accuracy mode, the bge-reranker-base model was deployed locally. When the server receives the query and retrieves the most relevant references, the model reranks the candidates, serving as a second filter on relevance. Specifically, the model performs pairwise comparisons and scores the texts within the candidate set against the user's query text. The scores are sorted to identify the top N of the most accurate answers, including the final references.

2.5. Evaluation Framework to Assess Model Performance. An evaluation framework was designed to assess the performance of the AMR-Policy GPT (Figure 2). First, the RAGAS framework was utilized for model optimization. Second, two sections of human evaluation were conducted, including independent evaluation and end-to-end evaluation. The independent evaluation assesses the performance of the AMR-Policy GPT in terms of information retrieval and answer generation within individual documents (each PDF/DOCX file of NAP was assessed one by one). The end-to-end evaluation evaluates the AMR-Policy GPT's performance in searching for information and generating answers across the entire database (all documents), with the comparison of different modes of the AMR-Policy GPT and OpenAI's ChatGPT (GPT-4).

Specifically, in the independent evaluation, a systematic manual analysis of NAPs was conducted using hybrid AMR-Policy GPT across a subset of countries in our database to ascertain the presence of key descriptions of budgetary allocations, appropriations, or funding strategies specific to addressing the AMR issue (see the Supporting Information, sheet "Independent Evaluation Data"). We chose this case as an example of AMR-Policy GPT performance because costing or financing components may be missing or unreliable, as they are not included in many NAPs.

The end-to-end evaluation using three sets of explicit experiments is detailed in Section 3.2 and Supporting Information (pages S16−S35). These experiments assessed answers to 10 progressively more complex questions on AMR using the standalone and hybrid AMR-Policy GPT modes and GPT-4, respectively.

2.6. Prompt Engineering. The hybrid and standalone versions of the AMR-Policy GPT use different reference sources. The hybrid mode integrates contextual references from the AMR policy database with in-house knowledge derived from GPT-4, refining the response based on wider insights. The detailed prompt engineering of the hybrid mode is shown in Figure 3. In contrast, the standalone mode solely retrieves contextual references from our own AMR policy database for responses. Both modes explicitly cite their sources, presenting referenced excerpts, document titles, publication years, and publisher information. In cases where queries cannot be adequately answered by the available database or the queries were potentially beyond the intended scope of the LLM and

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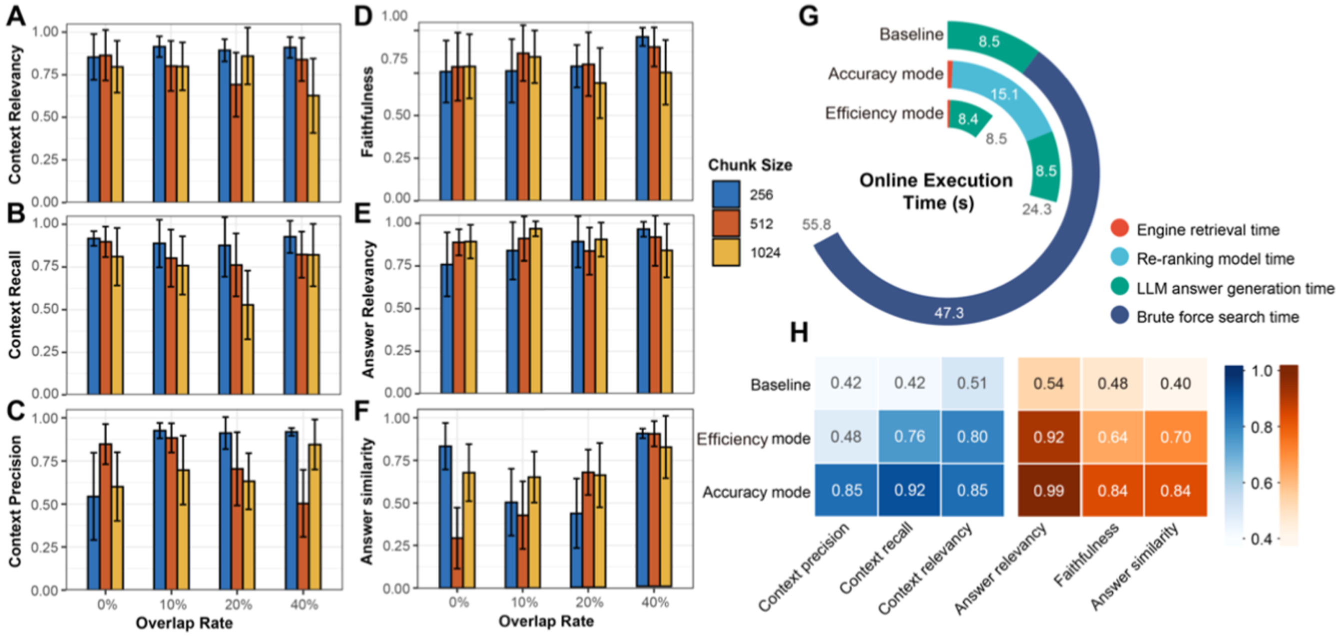

Figure 4. Model performance evaluation of AMR-Policy GPT. (A−F) RAGAS evaluation metrics (average answer relevancy, context recall, context precision, faithfulness, context relevancy, and answer similarity) of AMR-Policy GPT under different chunk sizes and chunk overlap rates. (G) Online execution time of accuracy mode and efficiency mode of AMR-Policy GPT and baseline model. (H) Heatmap of RAGAS evaluation metrics for accuracy mode and efficiency mode of AMR-Policy GPT, along with baseline model.

chatbot, a standardized response was provided: "there is not enough information".

3. RESULTS AND DISCUSSION

3.1. What Tool Is Needed to Assist AMR-Policy Guidance under One Health? NAPs for AMR have not always been developed employing cross-sectoral expertise and contributions, making many NAPs inadequate to support One Health-based solutions.[27] This is a big problem because it is commonly agreed that AMR will only be addressed through integrated action that spans the human, animal, crop, and environmental sectors.[28] In particular, many NAPs often state the importance of the environmental dimension of AMR in One Health planning but are then nonspecific about how environmental considerations will be operationalized or costed the follow-on action on environmental AMR surveillance rarely occurs, especially within national health protection programs. As a result, a decision-support tool that includes guidance and reports from all of the One Health sectors, including the environment, is needed to reduce this imbalance in NAP development. Specifically, an AI-based tool, which synthesizes information from the global "experience set", could assist policymakers, filling knowledge gaps with up-to-date information beyond their local or sectoral expertise.

The AMR-Policy GPT prototype presented here is a valuable starting point, especially for LMICs, which often lack data on which local AMR NAPs can be based. This guidance tool has immediate uses. First, the AMR-Policy GPT could provide information on existing multisectoral action plans and their implementation, including evidence that local policymakers can use for their purposes. However, it is critical to note that this tool is not designed to write a NAP for a user; i.e., it is an "intelligent" information source to extend the user's knowledge quickly. AMR-Policy GPT also has other uses, such as efficiently recruiting key cited references for researchers, broader public education, and updating information in existing NAPs as new research becomes available.

In subsequent sections, a comprehensive evaluation of AMR-Policy GPT's performance is provided, both in inquiries within individual documents and across the entire database, including comparisons with ChatGPT. A demonstration Road Test of the AMR-Policy GPT is also provided. However, we do not encompass here all types of inquiries or applications using the prototype (given the limited database as well as the diverse and limitless nature of users' questions) but aim to showcase the potential of AMR policy-focused LLM for future NAP guidance and wider research and educational purposes. Model performance of the RAG framework, answer quality, and referencing were first be assessed.

3.2. Model Optimization and Evaluation of Model Performance. The impact of the chunk size and overlap rate on model performance was assessed to determine the optimized parameter using the RAGAS framework. The chunk size represents a specific length of chunks split by the RAG system, which affects the retrieval accuracy. The overlap rate represents the ratio of shared tokens between two consecutive chunks, and it is used to ensure semantic coherence in the text. The experimental results (Figure 4A−C; detailed definitions of parameters are detailed in Supporting Information) indicate that larger chunk sizes have a negative impact on key indicators of retrieval performance (i.e., context recall, relevancy, and precision) across different overlapping rates. However, for generation module performance (shown in Figure 4D−F), larger chunk sizes enhance answer relevancy at lower overlap rates (zero and 10%), although as overlap rates increase, the lowest chunk size (256) model exhibited the highest answer relevancy. This implies that smaller chunks can lead to fragmented and less coherent content, thus requiring higher overlap to ensure answer relevancy. A model with a chunk size of 256 and an overlap rate of 40% provided the highest faithfulness and answer similarity, suggesting the minimization of "halluci-

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nation" effects under these conditions. Therefore, considering all performance indicators, a chunk size of 256 with an overlap rate of 40% was chosen for the prototype RAG system here.

Once the chunk size and overlap rate were chosen, the model's processing time in the accuracy and efficiency modes was assessed in comparison to the baseline model (see Figure 4G). The baseline model does not have an optimized search and retrieval algorithm, relying solely on brute force searches, a method that enumerates all potential candidates in an answer. Both the accuracy and efficiency modes outperformed the baseline model in all ways, which did not use the HNSW algorithm or reranking model. The baseline model took an average of 55.8 s to process a query and provide an answer, whereas the accuracy and efficiency modes took 24.3 and 8.5 s, respectively (Figure 4G). In terms of retrieval metrics, context relevancy of the accuracy (0.85) and efficiency modes (0.80) were 1.67 and 1.57 times better than the baseline model (0.51). Overall, the accuracy mode performed exceptionally well, surpassing the baseline model and the efficiency mode in terms of context recall and precision.

For generation module metrics, both the accuracy and efficiency modes achieved very high answer relevancy scores of 0.99 and 0.92, respectively. Additionally, the accuracy mode exhibited higher levels of context faithfulness and answer similarity compared with both the efficiency mode and the baseline model.

3.3. Evaluation of Answers. The next goal was to maximize the use of and value from the existing AMR policy document dataset. To improve data accessibility from extensive documents in an efficient and evidence-based manner, AMR-Policy GPT converts information from the dataset into dialogue. Two experiments were performed to assess the answer quality and AMR-Policy GPT performance. The first focused on individual documents, whereas the second evaluated AMR-Policy GPT's performance across the entire database. Both experiments involved query-based analysis and human evaluation.

3.3.1. Evaluation of Answers for Individual Documents. The current AMR database contains 160 NAP documents from 136 countries between 2011 and 2024. Some countries have different stages of NAPs (e.g., Norway, Canada, China; see Supporting Information), and some countries have multiple versions of NAPs, focusing on different perspectives. The current NAP database is in 27 languages, with English being the most prominent (99 documents), followed by French (18 documents) and Spanish (8 documents).

A cross-validation of NAP contents was performed between AMR-Policy GPT and manual reviews, and the results showed that AMR-Policy GPT had an accuracy rate of 87.26%, a precision rate of 97.59%, and a false positive rate of only 3.45% for information from individual documents (the record of performance is provided in the Supporting Information). As an example, AMR-Policy GPT consistently provided accurate and rational responses when queried about funding information in combating AMR. It is noteworthy that AMR-Policy GPT showed a high precision rate and a low false positive rate, which implies the guidance information provided by AMR-Policy GPT is trustworthy. Further, in scenarios with misleading or missing information, AMR-Policy GPT consistently responds with "there is not enough information" instead of providing an incorrect answer.

When unsuccessful answers of AMR-Policy GPT were examined, results showed that out of NAPs with funding information, there were 18 (11.46%) instances where AMR-Policy GPT failed to identify information. We suspect this is because NAPs sometimes present financial budget plans in tables using an image format, which the current model is unable to convert into chunks for processing. Variations in table layout, such as tables being oriented perpendicular to the regular text reading, also may have contributed to the model's inability to extract content. Conversely, the AMR-Policy GPT displayed strong performance in analyzing multilingual documents. We examined 58 NAPs written in non-English languages, and the AMR-Policy GPT effectively responded to queries from 54 of these NAPs. The model achieved an accuracy rate of 93.1% in locating and summarizing the relevant information, subsequently providing answers in English (the same language as the query) along with correct non-English citations.

However, further investigation is required to investigate the disparities that were observed during the assessments. For example, in AMR-Policy GPT development, 69 (43.95%) countries indicated they had a financial plan in their NAP. However, according to the 2023 TrACSS report,[29] only 20 out of 177 countries state that they have incorporated "financial provision for the National AMR action plan implementation···in the national plans and budgets". Twenty-nine other countries mention that their AMR NAPs have a budgeted plan for operation and monitoring mechanisms. These different responses suggest that while many countries have developed policies and laid out plans, their plans appear not to have been actioned.

3.3.2. Evaluation of Answers for the Entire Database. To evaluate the expertise, context-relevant, and accuracy of AMR-Policy GPT to search and integrate information across the entire database, we conducted three sets of human evaluation experiments by submitting a set of 10 questions to the standalone and hybrid AMR-Policy GPT bots and GPT-4 (Table 1). The difficulty of the questions ranged from level one to five (definitions of the difficulty level are provided in Supporting Information); some answers can be directly derived from the information available in the documents, while others require the integration and inference of evidence from database documents.

Table 1. Ten Designed Questions for Evaluating the Model Performance of the Standalone AMR-Policy GPT, Hybrid AMR-Policy GPT, and GPT-4[a]

questions	difficulty
what is AMR?	1
how many countries and areas were enrolled in GLASS-AMR?	2
how many countries have developed their own NAP on AMR?	2
which sector in One Health would be influenced by AMR?	3
are there Sustainable Development Goals (SDGs) indicators in addressing AMR?	3
which antimicrobial-resistant bacterial infections cause the highest mortality rates in Europe?	4
are there new emerging AMR pathogens or genes?	4
which countries monitor or plan to monitor the antimicrobial-resistance genes?	4
which regions will be disproportionally affected by AMR?	5
where is the majority of handling AMR finance going?	5

[a]For detailed information on the response of AMR-Policy GPT and ChatGPT, please refer to the Supporting Information. Additionally, for more examples illustrating the application of AMR-Policy GPT, please refer to the road test.

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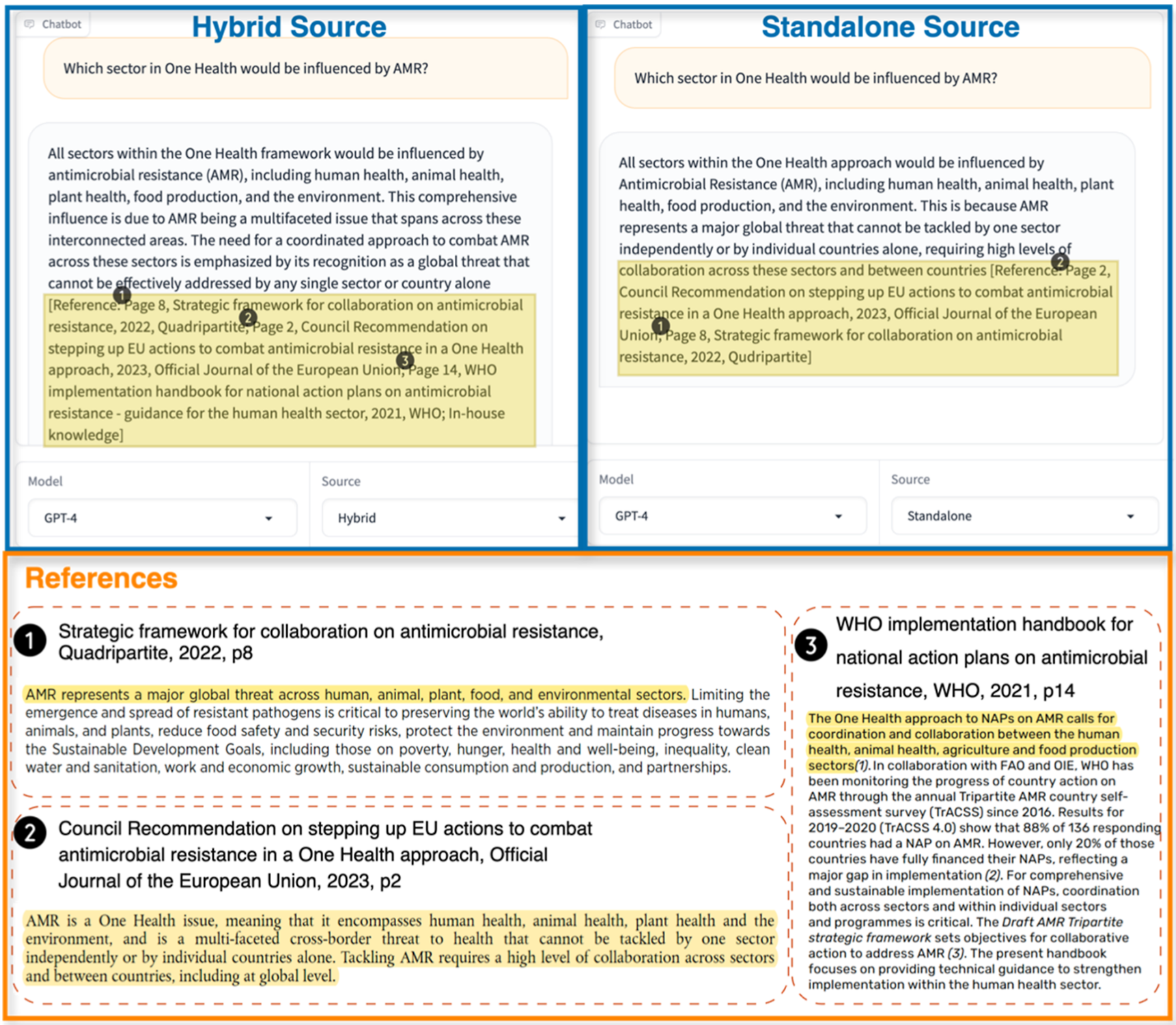

Figure 5. Evaluation of the reference. Upper part outlined in blue illustrates the answer platform of AMR-Policy GPT. Within this part, the left section represents the conversation derived from a hybrid source, integrating the AMR policy document database and LLM in-house knowledge, while the right section pertains solely to a standalone source, relying exclusively on our AMR policy document database. Lower part outlined in orange indicates the corresponding references in our database.

The quality of the responses varied dramatically. For example, GPT-4 only accurately responded to the question "what is AMR?" However, when confronted with more intricate inquiries, its responses lacked precision, provided overgeneralizations, or misinterpreted the question. For instance, when asked, "which countries monitor or plan to monitor AMR genes (ARGs)?" GPT-4 focused on AMR surveillance, not recognizing the question was specifically about gene monitoring, missing the keyword "genes". Answers also did not consider sectoral complexities, especially in environmental surveillance, or consider different detection methods that should be performed in parallel, which are critical in developing integrated AMR surveillance systems under One Health.[27]

Both the standalone AMR-Policy GPT and the hybrid AMR-Policy GPT provided much more informed and relevant answers than GPT-4, including key citations in their response sentences. GPT-4 often provided out-of-date answers from sources that have since been superseded by recent work. As for the question, "which sector in One Health would be influenced by AMR?" the GPT-4 answer did not include the plant health sector, which is a focus in the One Health Joint Plan of Action (2022–2026): working together for the health of humans, animals, plants, and the environment.[30]

Several factors contribute to this type of exclusion when using GPT-4. First, LLMs, such as GPT-4, lack current knowledge since they have been pretrained using historical data[17] (e.g., the pretraining data cutoff date for GPT-4 is April 2023). Moreover, such models serve large and general purposes and lack specific domain knowledge, which is key in more technical and nuanced AMR policy queries. Finally, GPT-4 often displays hallucinatory behaviors,[18] resulting in inaccurate answers. The impact of such problems is magnified relative to the environmental dimension of AMR, which is currently less standardized than in the human and animal healthcare sectors.

3.4. Evaluation of Response References. Using the developing model, we cross-checked the model answers to verify the accuracy of references provided by RAG systems (Figure 5). AMR-Policy GPT consistently cited correct sources, being accurate to the paragraph and page in their answers, which is important in validating the accuracy of the responses. The AMR-

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Policy GPT also showed a solid ability to generate concise summaries on specific content logically (a detailed analysis is provided in Supporting Information).

3.5. Overall Performance Analysis of AMR-Policy GPT. The AMR-Policy GPT prototype showed clear performance improvements under the RAG framework, especially in providing accurate and traceable document analysis. First, by leveraging external knowledge sources from a database, RAG empowers LLMs to incorporate specialized knowledge effectively. Based on the integration of external knowledge, the content generated by RAG exhibits a more coherent and logically structured narrative.

Second, RAG improves the contextual relevance of the generated text by integrating an external context. Consequently, RAG-generated responses more often align with the user's query or context, offering more meaningful and contextually appropriate answers. Hence, our model can serve diverse users, including researchers, policymakers, and stakeholders who seek science-informed and -integrated inquiries. The model also can serve the wider public, answering questions about AMR status, national initiatives, and strategies for preventing and controlling AMR. As such, it has great potential for public education via open and reliable knowledge sharing guided by official documents.

The model's responses are not only precise but also can be traced back to their respective sources (including source document names and page numbers), enhancing the interpretability and traceability of the content and reducing the probability of hallucinations. This aligns with users' requirements in policy development processes, particularly for expert AMR policymakers and stakeholders. They seek answers that require source attribution for the validation. Therefore, the tool will help to reduce the need for subsequent manual research and information retrieval, resulting in time and resource savings.

Additionally, the AMR-Policy GPT can handle multilingual queries and generate multilingual documents, making it suitable for international applications. This capability is particularly valuable in addressing AMR, which is a global health problem that extends across income settings, especially impacting lower-resource settings with diverse languages, such as parts of Africa and Asia. The model's multilingual capacity is intrinsically inclusive, which should help reduce perceived English-speaker bias in AMR guidance, promoting the One World concept in solving problems.[31] Further, with the increasing volume of related documents, RAG-based models demonstrate potential due to their high scalability, accommodating more diverse knowledge sources in developing AMR policy guidance documents. In parallel, the model can easily keep pace with new information as the external database is updated, which is planned via user feedback.

3.6. Limitations and Future Work. Although our study shows the potential of RAG-based LLMs for creating AMR policy guidance and documents based on related work from around the world, the methods have some limitations, and there are avenues for additional work. First, limitations in document accessibility prevented us from incorporating all global policy documents related to AMR on the publication date of this study (the cutoff date for the prototype was March 20, 2024). Additionally, although multilingual reports are included (e.g., French, German, and Chinese), our initial document search is performed in English, potentially introducing bias by limiting the number of collected documents. However, we will encourage users from diverse regions and linguistic backgrounds to share additional relevant documents, which will allow the AMR-Policy GPT to be updated and made more global. Further, a fraction of documents in our demonstration dataset was found to be invalid, often due to their format being in jpg or scanned copies. Therefore, input scrutiny is always needed. Finally, the AMR-Policy GPT currently does not support interpreting graphical data, which is under development for future versions.

Given the diverse components of AMR research, which necessitate a One Health approach, this field is ideally suited for leveraging LLM capabilities. On a general level, advancing scientific understanding to elucidate the epidemiology, burdens, drivers, and control mechanisms associated with AMR must be central in informing policy decisions. However, in most countries, developing NAPs is tasked to human health officials, animal health experts, or those who work in food safety, whereas growing evidence shows that AMR migrates through the environment, something often underestimated by other sectors. It is imperative to include operationalized information on environmental AMR development, transmission, and spread, as they can impact AMR exposure risks in other One Health sectors.

In the future, our goal is to integrate more harmonized and systematic surveillance data, reports, and scientific research from the environment and other sectors worldwide,[32] thereby enhancing the efficacy of the tool to assist in multisectoral policymaking. For example, a recent study fine-tuned GPTs with expert-level and closed-book wastewater engineering knowledge.[33] We hope lessons learned from using the AMR-Policy GPT will share knowledge across sectors and create more informed NAPs. Integrating scientific knowledge with policy documents that address real-world societal needs constitutes a crucial direction for all future scientific research endeavors.[34] However, extending the database to include published, peer-reviewed publications will require further work to evaluate potential biases from the inclusion or exclusion of articles based on copyright licensing, quality review, and other factors.

3.7. Using Answers from AMR-Policy GPT in Policy Guidance under One Health. The primary purpose of the AMR-Policy GPT is to assist users in searching for and integrating information effectively. However, we emphasize that it should not be directly used to generate policy. When policymakers utilize the AMR-Policy GPT, they should rely on their own insights and systematically search for relevant information within the framework they have established. They need to consider various factors, including specific information such as available funding and responsible departments, local circumstances such as prevalent pathogens and common infectious diseases in the area, as well as previous approaches employed to tackle similar AMR challenges. The information provided by the tool should be carefully reassessed to determine its relevance to the local circumstance and to evaluate its applicability. Failing to do so would result in the provision of generic and untargeted responses by AMR-Policy GPT, rendering them devoid of specificity or prioritization and ultimately meaningless.

For the application of AMR-Policy GPT in environmental implications, it can be utilized to retrieve information on current policy plans and actions in environmental sectors (Supporting Information, page S14), thereby offering references for addressing AMR mitigation. This is particularly valuable for LMICs where health and environmental protection infrastructure is limited. In such places, AMR is not addressed because local data are lacking and local policymakers do not

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know where to start in tackling such a massive and multifaceted problem. They could access information in sectors with constrained local surveillance systems by referring to One Health plans implemented in comparable nations or international agency publications without limitations imposed by accessibility to information resources and language barriers. If the settings are not comparable, then the policy evidence recruited may be less applicable, but the reference source would enable a policymaker to determine the relevance of the available evidence and evaluate it for local policymaking. However, it is important to highlight that while AMR-Policy GPT could serve as an effective tool to support policymaking through information sharing and knowledge integration, it should not be relied upon excessively. The tool is not comprehensive nor customized for every context, and therefore, it will require updates. This is, however, among our aspirations because shared knowledge in capacity development should enhance the model over time; e.g., more reliable sources from low-resource settings could be embedded in AMR-Policy GPT's knowledge base.

The future plan is to integrate scientific knowledge under different One Health sectors with policy information to create an enhanced AMR-Policy GPT. However, for the prototype here, we focused on the policy itself and provided a flexible RAG-based LLM framework for future expansion. We recognize that AMR and its complexity from various One Health sectors require delving far beyond policy documents, but as a starting point, the evidence here shows that the AMR-Policy GPT has demonstrable value, especially for places with very limited local information. On a grander level, the work here shows how LLMs can translate information, knowledge, and data into evidence important for AMR mitigation, hopefully promoting cross-sectoral discourse and collaboration among policymakers, researchers, and other relevant parties.

■ ASSOCIATED CONTENT

Data Availability Statement

All codes are provided at https://github.com/ShuleLi-IUE/AMRGPT.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.4c07842.

- Supporting methods and results; multi-language question and answer examples; comparison of generated answers from the hybrid AMR-Policy GPT, standalone AMR-Policy GPT, and GPT-4 to ten questions (PDF)
- Road test of AMR-Policy GPT (PDF)
- Model performance on RAGAS framework metrics and independent evaluation and detailed information of knowledge base and included NAPs (XLSX)

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Notes

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